

Block 3 — Chapter 9: Fluid Lines & Fittings

Plain-English Study Guide — read this instead of the whole chapter

amt-tutor make-up study guide · 2026-06-14 · 5th-grade-teacher level · feeds class + cumulative Block exam
· supports hangar projects G22/G23

Source (page numbers cited on every fact): FAA-H-8083-30 (Aviation Maintenance Technician — General), Chapter 9 "Fluid Lines & Fittings," pages 9-1 through 9-23. Citations like [\[p. 9-3\]](#) point straight to the page so you can double-check anything.

The one-sentence big idea: Airplanes move fluids (fuel, oil, hydraulic, oxygen, instrument air) through two kinds of plumbing — **rigid metal tubing** for long straight runs that hold still, and **flexible hose** wherever there is movement or vibration. This chapter is how you cut, bend, flare, fit, identify, install, and inspect both. [\[p. 9-1\]](#)

1. The Two Kinds of Plumbing (start here)

Rigid fluid lines = metal tubing. Used for stationary, long, relatively straight runs — fuel, oil, coolant, oxygen, instrument, and hydraulic lines. [\[p. 9-1\]](#)

Flexible hose is used with *moving* parts or anywhere there is a lot of vibration. [\[p. 9-1\]](#)

5th-grade picture: Rigid tubing is like the copper water pipes bolted inside your house walls — they don't move. Flexible hose is like the bendy hose under your kitchen sink that lets the faucet wiggle without snapping the pipe. On a plane, anything that shakes or moves gets the bendy hose; everything that holds still gets the stiff metal pipe.

Replacement rule: a replacement line must be the *same size and material* as the original. All tubing is pressure-tested before first install and is built to take several times normal operating pressure. If a tube bursts or cracks, it is almost always from **excessive vibration, improper installation, or collision/impact damage** — and you must find out *why* it failed, not just replace it. [\[p. 9-1\]](#)

2. RIGID Tubing — Materials & How to Tell Them Apart

Copper was used in the old days; modern aircraft use **aluminum alloy, corrosion-resistant steel (CRES), or titanium**. [\[p. 9-1\]](#)

Tubing materials and what each one is for

Material	Used for (pressure / job)	Notes & fittings
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1100 H14 & 3003 H14 (1/2-hard aluminum)	General-purpose lines, low or negligible pressure — instrument lines, ventilating conduits	The soft, low-pressure aluminums
2024-T3, 5052-O, 6061-T6 aluminum	Low & medium pressure — hydraulic/ pneumatic 1,000–1,500 psi , fuel & oil lines	The stronger aluminums
Steel — CRES 304, CRES 321, or CRES 304-1/8-hard (annealed)	High-pressure hydraulics (3,000 psi or more) — landing gear, flaps, brakes, fire zones, and anywhere foreign-object damage (FOD) is a risk (gear / wheel wells)	Higher strength = thinner walls, so weight isn't much more than aluminum. Uses swaged or MS flareless fittings
Titanium 3AL–2.5V	Transport-category & high-performance hydraulics, above 1,500 psi	30% stronger than steel, 50% lighter than steel. Uses Cryofit or swaged fittings

Exam trap / safety rule: NEVER use titanium tubing or fittings in any oxygen system. Titanium is oxygen-reactive — a fresh titanium surface exposed to gaseous oxygen can **spontaneously catch fire even at low pressure.** [p. 9-1]

Identifying which aluminum alloy you have — the color-code bands

You can spot aluminum vs. steel vs. titanium by eye, but you **cannot** eyeball which *aluminum alloy* it is. [p. 9-1] On **large** aluminum tubing the alloy number is **stamped** on the surface. On **small** tubing it's usually a **color-code band, no more than 4" wide**, painted at the two ends and about midway. If the band is two colors, each color gets half the width. [p. 9-1] If the code is unreadable, you **hardness-test** a sample to identify it. [p. 9-2]

Aluminum Alloy	Color of Band	Aluminum Alloy	Color of Band
1100	White	5052	Purple
3003	Green	6053	Black
2014	Gray	6061	Blue and Yellow
2024	Red	7075	Brown and Yellow

From Figure 9-1, p. 9-2. To positively match a replacement, compare its code markings against the original tubing you're removing. [p. 9-1]

3. Tubing SIZE — the OD rule (memorize this)

Rigid metal tubing is sized by **OUTSIDE DIAMETER (OD)**, measured in *sixteenths of an inch*. [p. 9-2]

So “number 6” tubing = $6/16" = 3/8"$ OD. “Number 8” = $8/16" = 1/2"$ OD. The OD is printed right on the tube.

You also need the **wall thickness** — printed on the tube in *thousandths of an inch*. To get the inside diameter (ID): **ID = OD – (2 × wall thickness)**. Example: a number 10 tube ($10/16" = 0.625"$ OD) with 0.063" walls has ID = $0.625" - 2(0.063") = 0.499"$. [p. 9-2]

Remember the contrast (it's a favorite test pairing): RIGID TUBE = sized by **OD**.
HOSE = sized by **ID** (covered in Section 9). Two different rules — don't mix them up.

4. Fabricating a Metal Tube Line — Cut, Bend, Flare, Bead

Tube forming is **four processes: cutting, bending, flaring, and beading**. Small soft tubing can be hand-bent during install; tubing **1/4" diameter or larger cannot be hand-bent** without tools. [p. 9-2]

4a. Cutting

Goal: a **square end, free of burrs**. Use a tube cutter (good for soft copper/aluminum) or a fine-tooth hacksaw. Special **chipless cutters** exist for hard 6061-T6, CRES steel, and titanium. [p. 9-2]

1. Cut the new piece about **10% longer** than the old one to allow for bending variations. [p. 9-2]
2. Put the tube in the cutter, cutting wheel on the mark; rotate around the tube, adding light pressure by twisting the thumbscrew a little at a time. **Too much pressure deforms the tube or makes heavy burrs**. [p. 9-2]
3. **Deburr** inside and outside with a knife, the cutter's burring edge, or a deburring tool (turn it end-for-end to do both). [p. 9-2]
4. File the end **square and smooth**. If using a hacksaw, prefer **32 teeth per inch** (a saw work-hardens the metal less than re-cutting). [p. 9-2]

Watch out: when deburring, do NOT thin or fracture the end wall. Even slight damage there leads to **fractured or defective flares that won't seal**. [p. 9-2]

4b. Bending — and why kinks/flat spots fail

Goal: a **smooth bend without flattening**. Tubing under 1/4" can usually bend by hand; larger sizes use portable hand benders or production benders. [p. 9-2] Standard bend radii vary with tube size (Figure 9-4). [p. 9-4]

The number to know: a little flattening in a bend is OK, but the flattened (small) diameter **must not be less than 75% of the original OD**. Tubing that is flattened, wrinkled, or has irregular bends **must not be installed**. [p. 9-3]

Why these fail: *Excessive flattening causes fatigue failure* of the tube. *Wrinkled bends* usually come from trying to bend thin-wall tubing without a bender. [p. 9-3] Four bend results to recognize (Figure 9-8): **Perfect** (good), **Flattened** (bad), **Wrinkled** (bad), **Kinked** (bad). [p. 9-3]

Using a hand bender: insert the tube in the groove so the measured end is left of the form block, align the two zeros, line the mark on the tube up with **L** (or **R** if the measured end is on the right), then pull the form handle steadily until the zero lines up with your desired angle on the radius block. Benders are size-specific — pick the right one for the tube diameter (size is stamped on the bender). [p. 9-3]

4c. Alternative bending method (no bender available)

Pack the tube with a **filler** so it can't collapse, then bend it over a forming block. [p. 9-3]

- **Dry-sand method:** plug one end, fill/pack with fine dry sand, plug the other end **tightly** (so plugs don't blow out), then bend over a block shaped to the radius. [p. 9-3]
- **Fusible-alloy method:** fill the tube under hot water with a **fusible alloy that melts at 160 °F**, cool it, bend slowly by hand around a block, then re-melt under hot water to remove the alloy. [p. 9-3]
- Either way: **remove ALL filler particles** and borescope-inspect so nothing gets carried into the system. Never heat the fusible alloy any other way than prescribed, or it sticks inside and ruins the tube. [p. 9-3]

5. FLARING — Single vs. Double (high-yield)

A **flare** is the funnel-shaped, flared-out end of a tube that a fitting clamps against to make a leak-proof seal. Two kinds: the **single flare** and the **double flare**. Flares see very high pressure, so the shape must be exact — **too small** = weak joint that leaks/pulls apart; **too large** = interferes with the

screw threads and leaks; **crooked** = the tube wasn't cut square. You cannot fix a bad flare by adding torque. [p. 9-3 / 9-4]

THE ANGLE TO MEMORIZE: aircraft flaring tools make a **35° to 37°** flare (call it the **37° standard**). **NEVER use an automotive 45° flaring tool** on aircraft tubing. [p. 9-4]

Single flare vs. Double flare — when to use each

	Single Flare	Double Flare
What it is	One flared-out funnel end	The end is belled out, then folded back on itself — two thicknesses
When used	General flaring	Soft aluminum-alloy tubing 3/8" OD and under
Why	Standard	Stops the soft flare from being cut off and failing under pressure; more concentric, seals better , resists the shearing effect of torque

MUST-KNOW flare rule

- **37° is the aircraft flare angle (35°–37°). Automotive 45° is forbidden.**
- **Double flare = soft aluminum alloy, 3/8" OD and under.**

5a. Rolling-type flaring tool instructions

Material limit: rolling-type tools are **only** for soft **copper, aluminum, and brass**. **Do NOT use on corrosion-resistant steel or titanium.** [p. 9-4]

1. Cut the tube squarely and remove all burrs. [p. 9-4]
2. Slip the **fitting nut and sleeve** onto the tube *first* (you can't add them after the flare). [p. 9-4]
3. Loosen the clamp screw, separate the die, and insert the tube in the die hole that matches its size. These tools are **self-gauging** — clamp the tube **flush with the top of the die block** for the correct flare. [p. 9-4 / 9-5]
4. Advance the clamp screw against the end segment, tighten firmly, move the yoke down over the die holder and twist clockwise to lock. [p. 9-5]
5. Turn the feed screw down firmly until you feel **slight resistance** — that means the flare is done. (Always read the tool's own instructions; types vary.) [p. 9-5]

5b. Double-flaring instructions

1. Deburr inside and outside; cut off the end if it looks damaged. [p. 9-4]
2. **Anneal** brass, copper, and aluminum first — heat to a **dull red** and cool rapidly in cold water (this softens the metal so it folds without cracking). [p. 9-4]
3. Open both clamping screws, pick the flaring-bar hole that matches the tube, and set the tube so it extends above the bar by the thickness of the adapter shoulder; tighten the clamps. [p. 9-4]
4. Insert the pilot of the correctly sized **adapter** into the tube, slip the yoke over the bars and center it. [p. 9-4]
5. Advance the cone down until the adapter shoulder rests on the bar — this **bells out** the end. [p. 9-4]
6. Back the cone off, remove the adapter, then advance the cone **directly into the belled end** — this **folds the tube on itself** into a double flare. **Don't overtighten** or you thin the flare wall. [p. 9-4]

5c. Beading (the fourth forming process)

A **bead** is a raised ring rolled around the tube near the end so a hose can slip over it and a clamp can grip without sliding off. Done with a hand beading tool (for **1/4" to 1" OD**), machine rolls, or grip dies; lubricate inside and outside with light oil. Beading machines are usually for large tubing over 1 15/16". [p. 9-7]

6. Fluid Line IDENTIFICATION — color codes, words, symbols

Fluid lines are marked so anyone can instantly read each line's **function, contents, and primary hazard** using **color codes, words, and geometric symbols** (Figure 9-16). [p. 9-7]

How markers are applied:

- Usually **1" tape or decals**. [p. 9-7]
- **Steel tags** instead of tape on lines **4" diameter or larger**, oily-environment lines, hot lines, and some cold lines. [p. 9-7]
- **Paint** in engine compartments — so tapes/decals/tags can't be sucked into the engine induction system. [p. 9-7]
- Markers go on **both ends** of a line, at least once in each compartment a line passes through, and right next to every valve, regulator, filter, or accessory. [p. 9-7]

The three hazard words (memorize what each means):

- **FLAM** = flammable — fuel lines. [p. 9-7]

- **TOXIC** = lines with toxic materials (replaces FLAM). [p. 9-7]
- **PHDAN** = PHysically DANGerous — oxygen, nitrogen, Freon. [p. 9-7]

Who is responsible: the aircraft/engine **manufacturer** does the original markers; the **mechanic** is responsible for **replacing** them when needed. [p. 9-7]

7. Fluid Line END FITTINGS

Fittings join tubing to a unit, to another tube, or to a hose. Two thread types: [p. 9-7]

Pipe threads	Machine threads
Tapered internal & external; the taper itself forms the seal (like household plumbing). Use approved thread lubricant to prevent seizing & leaks — but keep lubricant out of the system.	No taper, no sealing ability (like a nut and bolt). Only pull connections together or attach through bulkheads. Seal is made by a flared connection, crush washer, or synthetic seal . Size given in dash numbers = OD in 16ths of an inch .

Oxygen safety: do NOT use lubricants on oxygen lines — oxygen reacts with petroleum products and can ignite. Special oxygen-safe lubricants exist. [p. 9-7]

Bulkhead fitting: when a line passes through a bulkhead and must be secured to it, the bulkhead-fitting end is **longer** so a locknut can clamp it to the bulkhead. [p. 9-8]

The four fitting types (pressure & material decide which): (1) **bead & clamp** — low/medium pressure only (vacuum, coolant); (2) **AN flared**; (3) **MS flareless**; (4) **permanent** (Permaswage™, Permalite™, Cryofit™). Flared, flareless, and permanent types work at **any pressure**. [p. 9-8]

7a. AN flared fittings

A sleeve + nut; the nut draws the sleeve and the tube's flare tight against a cone-shaped male fitting to seal. The sleeve also supports the tube so vibration doesn't concentrate at the flare edge. [p. 9-8]

Color code: AN **steel** = **black**, AN **aluminum** = **blue**, aluminum-bronze = cadmium-plated/natural. [p. 9-8]

AN flared vs. MS flareless are NOT interchangeable. Tell them apart by the end: **AN flared has a CONE** at the end; **MS flareless has a STRAIGHT end**. [p. 9-9 / 9-8]

8. MS FLARELESS Fittings — the sleeve, presetting & tightening

MS flareless fittings are made for **high-pressure (3,000 psi) hydraulics** with severe vibration or fluctuating pressure. They need **no flaring**. Three parts: a **body**, a **sleeve (ferrule)**, and a **nut**.

[p. 9-9]

How it seals (5th-grade picture): as you tighten the nut, the body squeezes the **sleeve** so its sharp edge **bites into the outside of the tube** like a tiny ring of teeth grabbing the pipe. That bite is the seal. The body's counterbore has a reverse angle (**15° for steel** connectors, **45° for aluminum**) that stops the tube from collapsing inward and adds sealing force. [p. 9-9]

8a. PRESETTING a new flareless assembly (done before installation)

1. Cut the tube to length with **perfectly square ends**; deburr inside and out. Slip on the **nut first, then the sleeve**; lubricate the fitting and nut threads with hydraulic fluid. [p. 9-5/9-6]
2. Put the fitting in a vise. Hold the tube **firmly and squarely, bottomed in the fitting**. Tighten the nut until the sleeve's cutting edge grips the tube — to find that point, **slowly turn the tube back and forth** while tightening; **when the tube no longer turns, the nut is ready for final tightening**. [p. 9-6]
3. **Final tightening (memorize these turn counts):**
 - Aluminum-alloy tubing **up to and including 1/2" OD** → **1 to 1-1/6 turns**.
 - Steel tubing, and aluminum-alloy **over 1/2" OD** → **1-1/6 to 1-1/2 turns**. [p. 9-6]

After presetting, check: the tube must extend **3/32" to 1/8"** beyond the sleeve pilot (or blowoff can occur). Sleeve pilot must contact the tube or have at most **0.005" clearance for aluminum / 0.015" for steel**. Slight tube collapse at the sleeve cut is OK; **no movement of the sleeve pilot except rotation** is allowed. [p. 9-6]

8b. Installing a flareless fitting (in the aircraft)

1. Run the nut down **by hand** until resistance increases. If fingers can't do it, use a wrench but watch for **first signs of bottoming**. [p. 9-12]
2. Final tightening starts where the nut **just begins to bottom**. Then turn the nut **one-sixth turn (one flat on the hex nut)**, holding the connector still with a second wrench. [p. 9-12]
3. Pressure-test. If it leaks, you may tighten an **additional one-sixth turn (one-third turn total max)**. [p. 9-12]
4. Still leaking after one-third turn? **Remove and inspect** for scores, cracks, foreign material, or overtightening damage. [p. 9-12]

CAUTION: Never tighten a flareless nut beyond one-third turn (two flats). That's the maximum before you permanently damage the sleeve and nut. Overtightening drives the

sleeve's cutting edge too deep, weakening the tube so that **normal in-flight vibration can shear it.** [p. 9-14]

9. Other Permanent Fittings (quick hits)

- **Swaged (Permaswage™):** permanent, near maintenance-free; joins hydraulic lines where you won't routinely disconnect. Often used with **titanium and CRES.** Installed with portable hydraulic tooling; to remove, **cut the tube.** [p. 9-9]
- **Permalite™:** mechanically attached by **axial swaging** — a ring is moved along the fitting to deform it into the tube. [p. 9-9]
- **Cryofit:** a standard fitting with a **shape-memory-alloy (Tinel™)** sleeve. Made 3% small, frozen in liquid nitrogen, expanded 5% larger; slipped on, it **warms in 10–15 seconds, shrinks back, and bites the tube** for a permanent seal. Remove only by cutting the tube. Common with **titanium.** [p. 9-9/9-10]

10. Installing & Inspecting RIGID Tubing

Before install: inspect the line, remove dents/scratches, make sure nuts and sleeves are snugly mated and properly flared, and that the line is clean and free of foreign matter. [p. 9-10]

Three install rules the FAA loves to test:

- **Never put compound on the fitting faces or the flare** — it destroys the metal-to-metal contact that makes the seal. [p. 9-10]
- **Align the line BEFORE tightening. Do not pull the assembly into place with torque on the nut.** [p. 9-10]
- **Use a torque wrench and the correct torque.** Overtightening cuts off the flare or ruins the sleeve/nut; undertightening lets the line blow out or leak. [p. 9-10/9-12]

Why bends matter on install: never pick a path with **no bends.** A tube can't be cut/flared accurately enough to fit straight without strain, and bends are needed to let the tube **expand/contract with temperature and absorb vibration.** Use the old tube (or a hand-bent **wire template**) as a pattern, and start bends a reasonable distance from fittings so the nuts and sleeves can slide back. [p. 9-15/9-16]

Inspection & repair limits:

- Aluminum scratches/nicks **not deeper than 10% of wall thickness** (and not in the heel of a bend) can be burnished out by hand. [p. 9-14]
- A **dent under 20% of tube diameter** is acceptable (unless it's in the heel of a bend). Remove dents by drawing a **bullet** (ball/slug) through the tube. [p. 9-14]

- **Any crack or deformity in a flare = reject.** Replace lines with severe die marks, seams, or splits. [p. 9-14]
- Hard, thin-wall CRES and titanium have **much smaller** damage limits — consult the aircraft maintenance manual. [p. 9-14]

Support spacing: all fluid lines must be supported at set intervals (Figure 9-43 gives max distance between supports by OD & material — e.g., 1/4" aluminum every 13.5", same steel every 16"). [p. 9-23] Use **bonded clamps** for metal hydraulic/fuel/oil lines (remove paint/anodizing under the bonding clamp); unbonded clamps are for wiring only. [p. 9-23]

11. FLEXIBLE Hose — Materials & Construction

Pure rubber is never used. Synthetics give the needed strength, durability, and workability. The common ones: **Buna-N, neoprene, butyl, EPDM, and Teflon™.** Teflon is in its own category; the rest are synthetic rubbers. [p. 9-16]

Material	Good with	Do NOT use with
Buna-N	Excellent with petroleum products	Phosphate-ester fluid (Skydrol™). Don't confuse with Buna-S.
Neoprene (acetylene base)	Petroleum (not as good as Buna-N) but better abrasion resistance	Skydrol™
Butyl (from petroleum raw materials)	Excellent with Skydrol™ (phosphate-ester)	Petroleum products
Teflon™ (tetrafluoroethylene resin)	Nearly everything — fuel, petroleum/synthetic oils, alcohol, coolants, solvents. Temp range –65°F to +450°F ; almost unlimited shelf/service life	(broadest compatibility of all)

5th-grade memory hook: Buna-N and Neoprene love oil/gas but *hate* Skydrol. Butyl is the opposite — it loves Skydrol but *hates* oil/gas. Teflon is the easy-going one that gets along with almost everything.

Construction of rubber hose: a **seamless synthetic-rubber inner tube**, wrapped in layers of **cotton braid and wire braid**, with an outer layer of **rubber-impregnated cotton braid**.

[p. 9-16] **Teflon hose** is extruded into a tube then covered with **braided stainless-steel wire** for strength. [p. 9-17] Brake hoses sometimes get an extra stainless wire braid over the outside for protection. [p. 9-16]

11a. Rubber hose pressure classes

Class	Pressure	Reinforcement
Low pressure	below 250 psi	Fabric braid
Medium pressure	up to 3,000 psi (smaller sizes to 3,000; larger sizes to 1,500)	One wire braid
High pressure	all sizes up to 3,000 psi	Multiple wire braids

Pressure classes from p. 9-16. [p. 9-16]

12. Hose IDENTIFICATION & SIZE (the ID rule)

Printed on the hose is a **lay line** (a stripe running straight down the length) plus the **type, the quarter and year of manufacture, and a 5-digit manufacturer code**. The lay line marks the hose's **natural lay (no twist)** and is repeated at intervals of **no more than 9 inches**. [p. 9-16/9-17]
Hose suitable for phosphate-ester fluid is marked "**Skydrol use.**" [p. 9-17]

THE HOSE SIZE RULE (memorize the contrast): Hose is sized by a **dash number** that matches the tubing size it works with. The dash number is keyed to **INSIDE DIAMETER (ID)** — it does **NOT** denote OD. When the hose dash number equals the tube dash number, you have the right size. [p. 9-19]

Compare: **RIGID TUBE = OD in 16ths** [p. 9-2] vs. **HOSE = ID, by dash number** [p. 9-19]

13. Inspecting, Fabricating & Replacing Flexible Hose

13a. Inspection — signs of deterioration

At every inspection, look for: **leakage, separation of the cover/braid from the inner tube, cracks, hardening, lack of flexibility, or excessive "cold flow."** Any of these = replace the hose. [p. 9-18]

Define “cold flow”: the **deep, permanent impressions** left in a hose by the pressure of hose clamps or supports. Because of cold flow (a setting process), **recheck/retighten clamps for several days after install.** [p. 9-18 / 9-23]

- Hose with **swaged (factory) end fittings** that fails → **replace the entire assembly** (order new, correct size/length, factory fittings). [p. 9-18]
- Hose with **reusable end fittings** → you can **fabricate a replacement** using the manufacturer’s tooling/instructions. [p. 9-18]

13b. Fabricating a hose assembly (MS detachable fittings)

MS-type end fittings are **detachable and reusable** if serviceable. The fitting’s ID equals the hose ID. [p. 9-18] Steps (Figure 9-35):

1. Vise the hose; cut to length with a fine-tooth hacksaw or cutoff wheel. [p. 9-19]
2. (High-pressure hose only) slit the cover to the wire braid with a knife — don’t nick the braid — and twist the cover off with pliers. [p. 9-19]
3. Screw the **socket onto the hose counterclockwise.** [p. 9-19]
4. Liberally lubricate inside the hose and the nipple threads. [p. 9-19]
5. Screw the nipple into the socket; leave **0.005" to 0.031" clearance** between the nipple hex and the socket. [p. 9-19]

CAUTION on lubricant: for hose made for **synthetic fluids (Skydrol™ / HYJET)**, **do NOT use any petroleum product** as the assembly lubricant — use a **vegetable-soap liquid** instead. [p. 9-19]

Proof-testing: every hose must be **proof-tested after assembly** with internal pressure. Liquid (hydraulic oil/water) for hydraulic/fuel/oil; dry oil-free air or nitrogen for air/instrument lines (gas tests are done **underwater**). A repaired/overhauled hose should be tested to **at least 1.5× system pressure** before install. [p. 9-19]

14. Installing Flexible Hose Assemblies (projects G22/G23)

This is the hands-on part for the hangar. Four things to get right: SLACK, DO-NOT-TWIST, BEND RADIUS, and SUPPORT/CLAMPS.

- **SLACK:** never install hose with a mechanical load on it. Leave **5% to 8% slack/bend of total length** — pressurized hose **contracts in length and expands in diameter.** [p. 9-21]

- **DO NOT TWIST:** install without twisting (twisting can rupture the hose or loosen the nuts). **Use the lay line / linear stripe** to confirm no twist — a straight stripe means no twist; if it spirals, the hose is twisted. Swivel connections relieve twist stress. [p. 9-21 / 9-17]
- **BEND RADIUS:** never bend below the hose's **recommended minimum bend radius** — too-sharp bends drop the burst pressure far below rated. Use elbow fittings to relieve sharp bends. [p. 9-21]
- **FLEX / straight ends:** the hose must stay **straight for at least two hose diameters from the end fittings** (the metal ends don't flex). [p. 9-21]
- **SUPPORT:** support hose **at least every 24 inches** (closer is better); never stretch hose tightly between two fittings; keep clearance from all other lines/structure. [p. 9-21]
- **Preformed hose: never straighten a preformed hose assembly;** use a support wire if it must be removed. [p. 9-17/9-18]

14a. Hose CLAMPS & support clamps

Use a **hose-clamp torque-limiting wrench** when available (made in **15 and 25 in-lb** limits). With no torque wrench, use the **finger-tight-plus-turns** method (values are approximate — use judgment). [p. 9-22]

Situation	Tighten to
Initial install — self-sealing hose	~ 15 in-lb
Initial install — all other aircraft hose	~ 25 in-lb
Retighten in service — non-self-sealing	If clamp won't finger-tighten, leave it alone unless leaking; if leaking, tighten 1/4 turn
Retighten in service — self-sealing	If looser than finger-tight, bring to finger-tight and add 1/4 turn

Support clamp types: rubber-cushioned clamps secure lines subject to vibration (cushion stops chafing); **plain** clamps for non-vibration areas; **Teflon-cushioned** clamps where Skydrol/hydraulic fluid/fuel would attack rubber (but Teflon damps vibration less). [p. 9-22] Clamps smaller than the hose OD can **restrict flow** — use the correct size. [p. 9-23]

MUST KNOW — the verbatim test items for the Block exam

1. **37° flare (35°–37°) for aircraft tubing. Automotive 45° flaring tools are forbidden.** [p. 9-4]

2. **RIGID TUBE** is sized by **OUTSIDE DIAMETER (OD)**, in 16ths of an inch (No. 6 = 3/8"). **HOSE** is sized by **INSIDE DIAMETER (ID)**, by dash number. [p. 9-2 / 9-19]
3. **DOUBLE FLARE** is used on soft aluminum-alloy tubing 3/8" OD and under. [p. 9-4]
4. **Hazard markings:** **FLAM** = flammable (fuel); **TOXIC** = toxic; **PHDAN** = physically dangerous (oxygen/nitrogen/Freon). [p. 9-7]
5. **NEVER** use titanium in any oxygen system (oxygen-reactive, can ignite). [p. 9-1]
6. **Bend flattening:** small diameter of a flattened bend must not be less than 75% of original OD. [p. 9-3]
7. **Flareless tube install:** final tighten 1/6 turn from bottoming; never exceed 1/3 turn total (two flats). [p. 9-12/9-14]
8. **Flareless presetting:** aluminum $\leq 1/2"$ OD = 1 to 1-1/6 turns; steel & aluminum $> 1/2"$ OD = 1-1/6 to 1-1/2 turns. [p. 9-6]
9. **Hose slack** 5–8%; support every 24 in; lay-line stripe = do not twist; straight for 2 hose diameters from end fittings. [p. 9-21]
10. **Aluminum color codes:** 1100 White, 3003 Green, 2024 Red, 5052 Purple, 6061 Blue+Yellow, 7075 Brown+Yellow. [p. 9-2]

15. Quick Glossary (define-on-first-use, all in one place)

- **Rigid line / tubing** — stiff metal pipe for still, straight runs. [p. 9-1]
- **Flexible hose** — bendy line for moving/vibrating spots. [p. 9-1]
- **OD / ID** — outside / inside diameter. $ID = OD - 2 \times \text{wall}$. [p. 9-2]
- **Flare** — funnel-shaped flared tube end that seals against a fitting. [p. 9-3]
- **Bead** — raised ring rolled near a tube end so a hose+clamp grips it. [p. 9-7]
- **Sleeve / ferrule** — the ring in a fitting that bites or supports the tube. [p. 9-9]
- **Presetting** — pre-biting the sleeve onto a flareless tube before install. [p. 9-5]
- **Anneal** — heat metal to dull red & quench to soften it (before double-flaring). [p. 9-4]
- **Lay line** — the straight printed stripe that proves the hose isn't twisted. [p. 9-16/9-21]
- **Cold flow** — permanent dents in hose from clamp/support pressure. [p. 9-18]
- **CRES** — corrosion-resistant (stainless) steel. [p. 9-1]
- **FOD** — foreign object damage. [p. 9-1]

This is a plain-English summary of FAA-H-8083-30 Chapter 9 (pp. 9-1 to 9-23) for make-up study. Numbers, turn-counts, and limits are quoted from the handbook with page cites. For any graded AIM worksheet, match the answer to the AIM slide wording if it differs; this guide uses FAA handbook wording, which is what the written exam grades against.